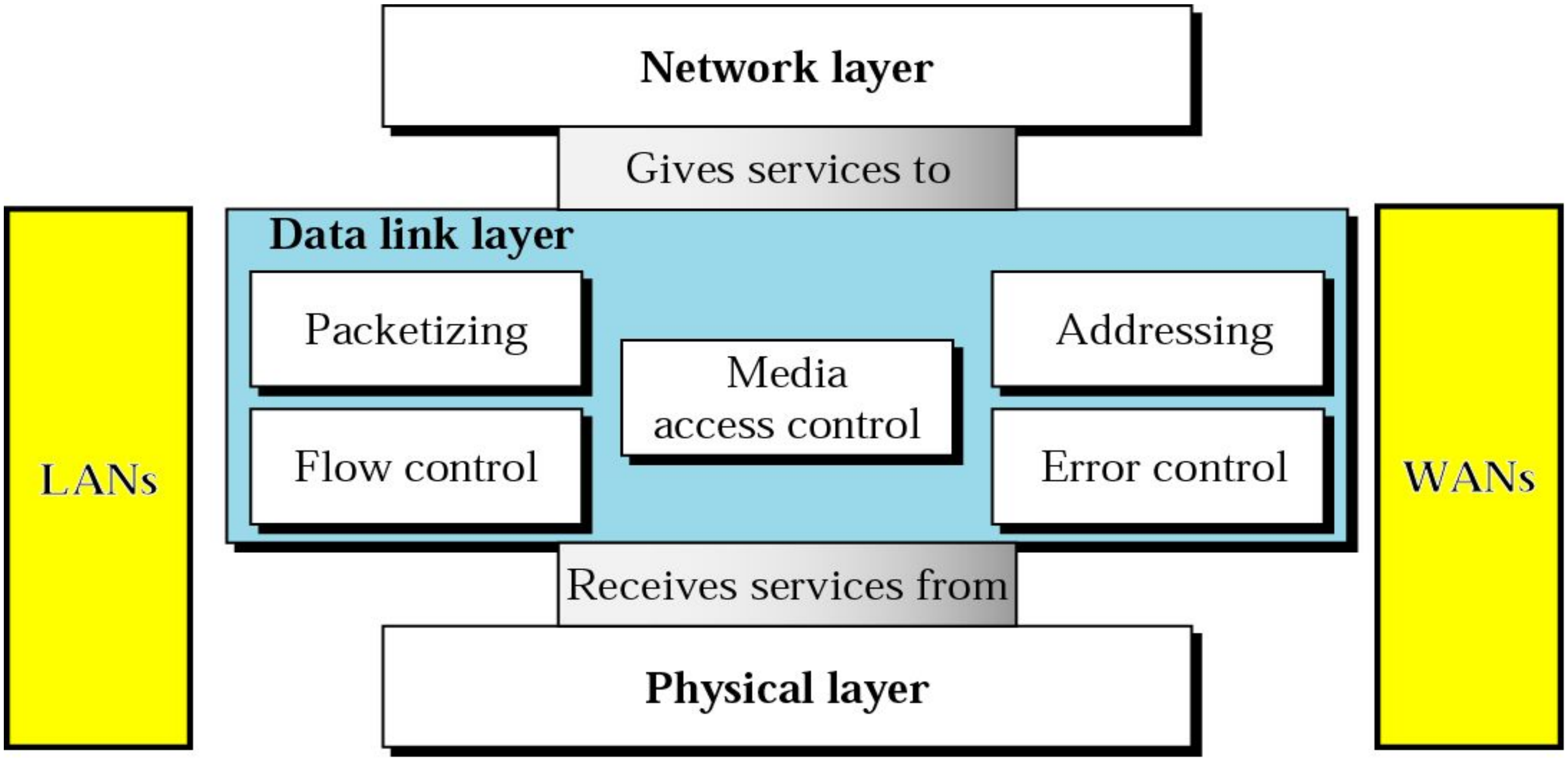
Computer Communication & Networks

Lecture 18

Datalink Layer: Local Area Network

Data Link Layer



Data Link Layer Topics to Cover

Error Detection and Correction

Data Link Control and Protocols

Multiple Access

Local Area Networks

Wireless LANs

IEEE Standards

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.

IEEE 802 Series of LAN Standards

- 802 standards free to download from http://standards.ieee.org /getieee802

IEEE 802®: Overview & Architecture

IEEE 802.1 ™ Bridging & Management

IEEE 802.2 ™: Logical Link Control

IEEE 802.3 ™: CSMA/CD Access Method

IEEE 802.4 ™: Token-Passing Bus Access Method

IEEE 802.5 ™: Token Ring Access Method

IEEE 802.6 ™: DQDB Access Method

IEEE 802.7 ™: Broadband LAN

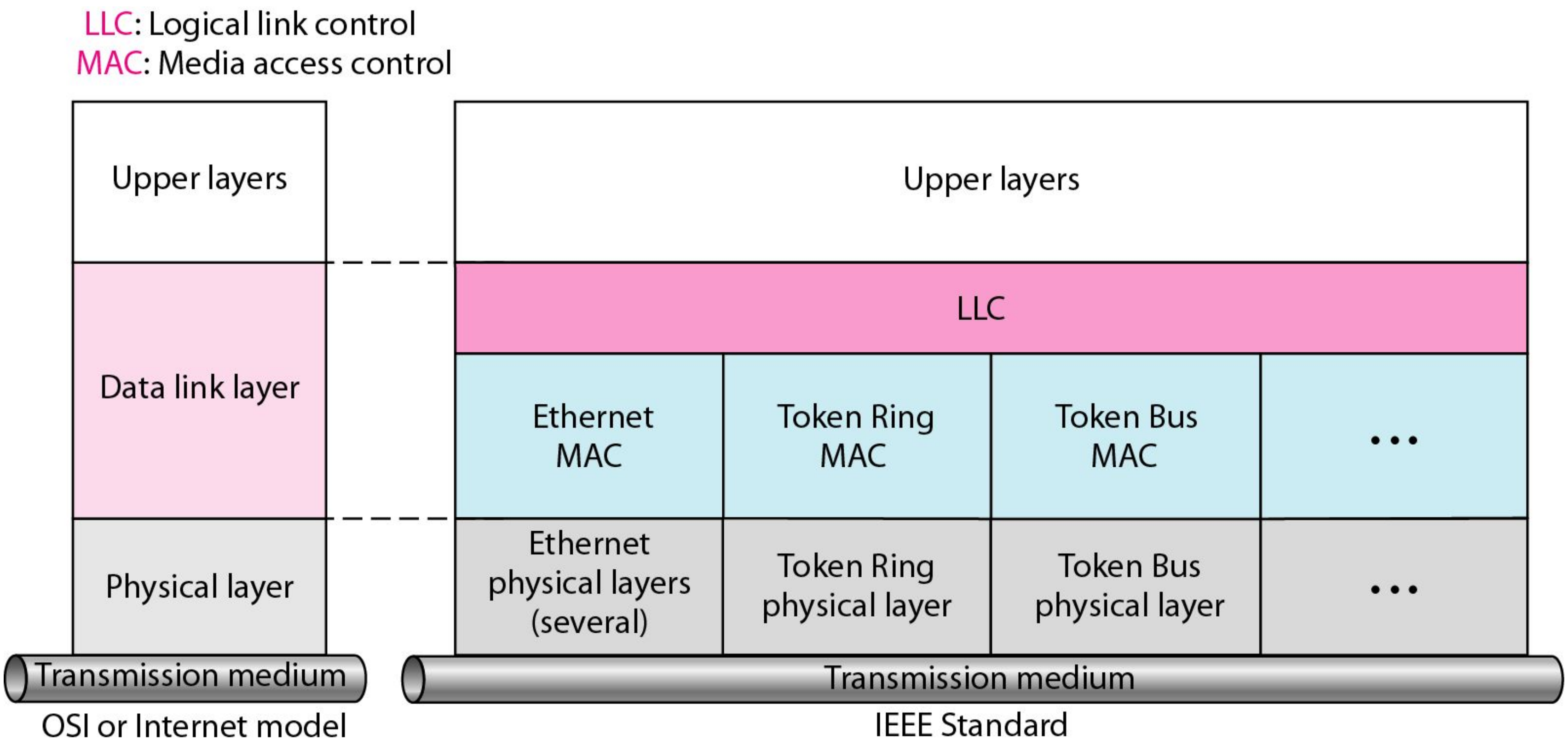
IEEE 802.10 ™: Security

IEEE 802.11 ™: Wireless

IEEE 802.12 ™: Demand Priority Access

WiMAX → IEEE 802.16™: Broadband Wireless Metropolitan Area Networks

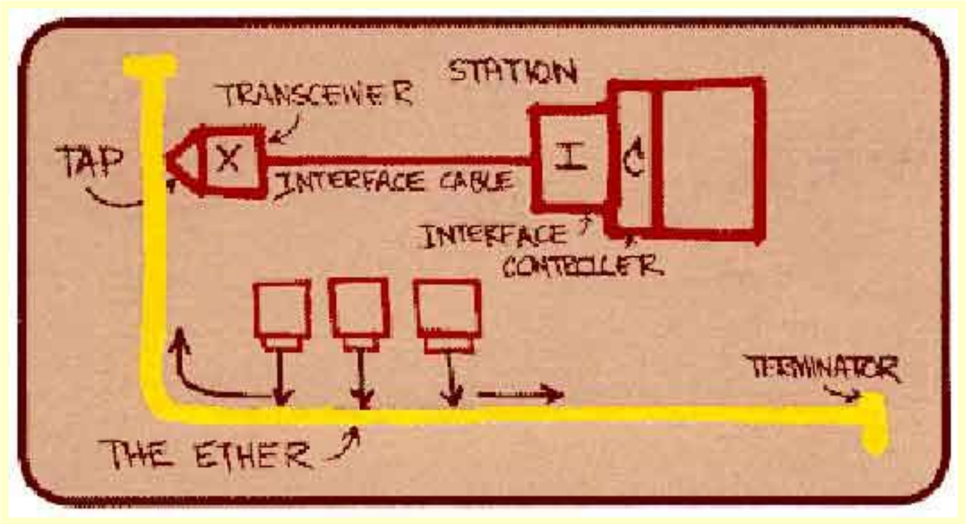
IEEE standard for LANs



Ethernet

"Dominant" LAN technology:

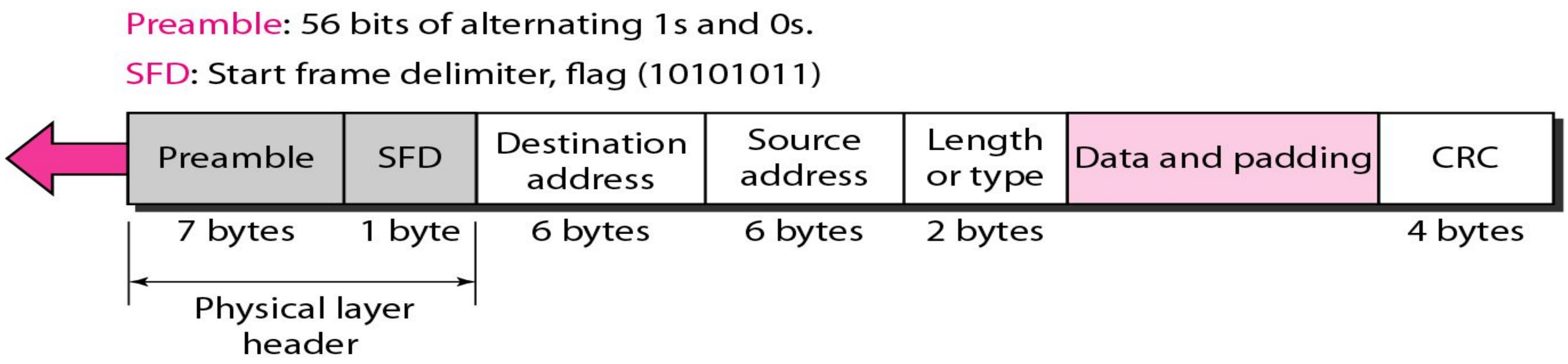
- Cheap $20 for 100Mbs!
- First widely used LAN technology
- Simpler, cheaper than token LANs and ATM
- Kept up with speed race: 10, 100, 1000 Mbps



Metcalfe's Ethernet sketch

Ethernet Frame Structure - 1

Sending adapter encapsulates IP datagram (or other network layer protocol packet) in Ethernet frame

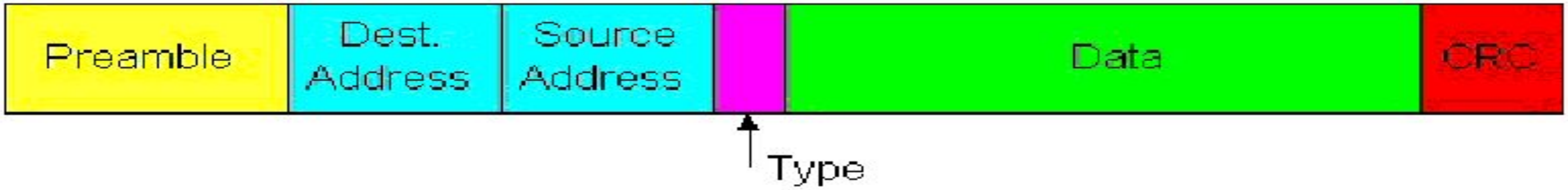


Preamble:

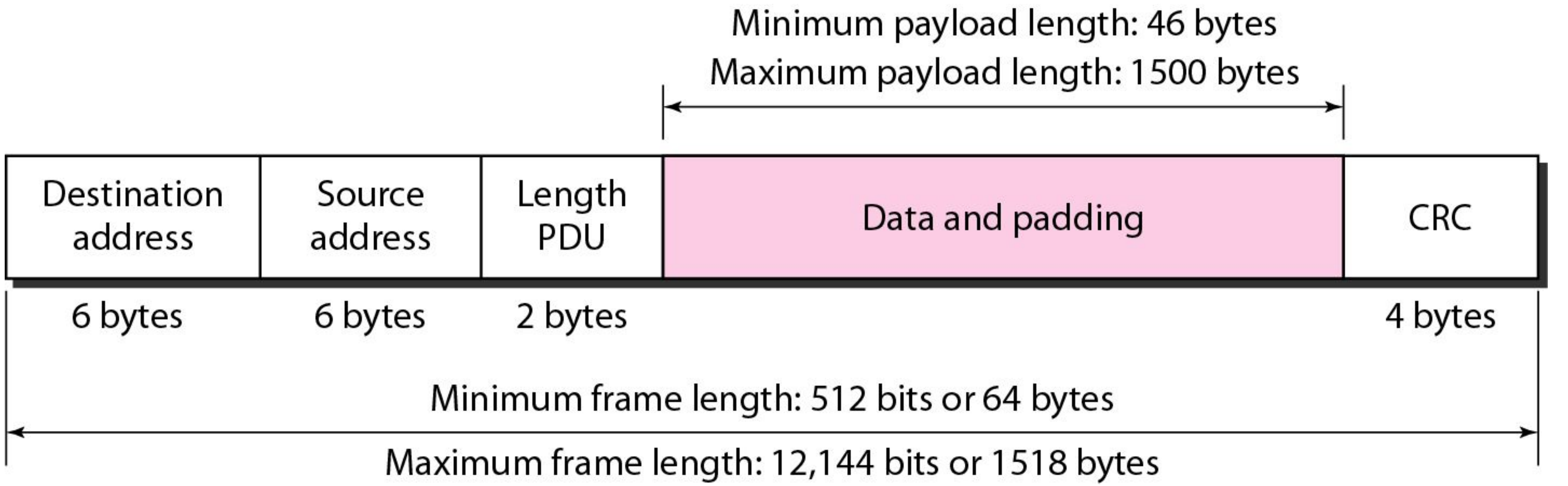
- **7 bytes with pattern 10101010 followed by one byte with pattern 10101011**
- **Used to synchronize receiver, sender clock rates**

Ethernet Frame Structure - 2

- **Addresses:** **6 bytes, frame is received by all adapters on a LAN and dropped if address does not match**
- **Type:** **indicates the higher layer protocol, mostly IP but others may be supported such as Novell IPX and AppleTalk)**
- **CRC:** **checked at receiver, if error is detected, the frame is simply dropped**



Minimum and Maximum Lengths



- Why there is an upper limit!!

Ethernet

- **Ethernet uses 1-persistent CSMA/CD on coaxial cable at 10 Mbps (802.3 allows other speeds & media)**
- **The maximum cable length allowed: 500m**
- **Longer distances covered using repeaters to connect multiple "segments" of cable**
- **No two stations can be separated by more than 2500 meters and 4 repeaters**
- **Including the propagation delay for 2500m and the store and forward delay in 4 repeaters, the maximum time for a bit to travel between any two stations is τ_{max}=25.6μse (one way)**

Ethernet: uses CSMA/CD

```
A: sense channel, if idle
   then {
          transmit and monitor the channel;
        If detect another transmission
         then {
            abort and send jam signal;
           update # collisions;
           delay as required by exponential backoff algorithm;
           goto A
           }
        else {done with the frame; set collisions to zero}
     }
   else {wait until ongoing transmission is over and goto A}
```

Ethernet's CSMA/CD

- **In order to ensure that every collision i s "heard" by all stations, when a station detects a collision , it jams the channel for**
- **Example**
  - **Two stations, A and B, are close together**
  - **A third station, C, is far away**
  - **A and B will detect each other's transmission very quickly and shut off**
  - **This will only cause a short blip which may not be detected by C but will still cause enough errors to destroy C's packet**

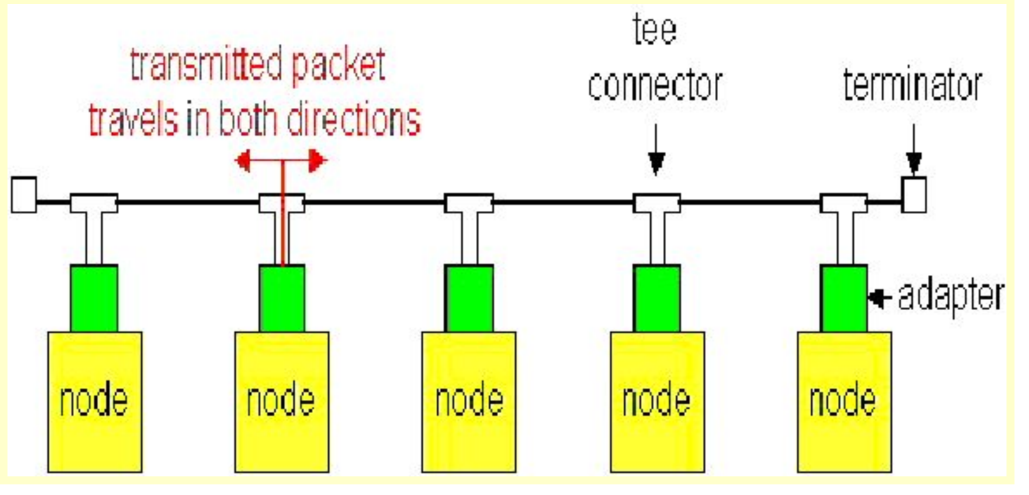
Ethernet's CSMA/CD

- When collisions occur, Ethernet uses a random retransmission scheme called exponential backoff:
  1. If your packet is in a collision, set K=2
  2. Pick a number k at random from {0, 1,..,K-1}
  3. After τ_{max} seconds, sense channel, transmit if idle
  4. If collision occurs, let K=2 x K, go to step 2

- After 10 repeats, stop doubling K
- After 16, give up and tell layer above "I give up"
- "Fixes" random access stability problem by passing it to the layer above!

Ethernet Technologies: 10Base2

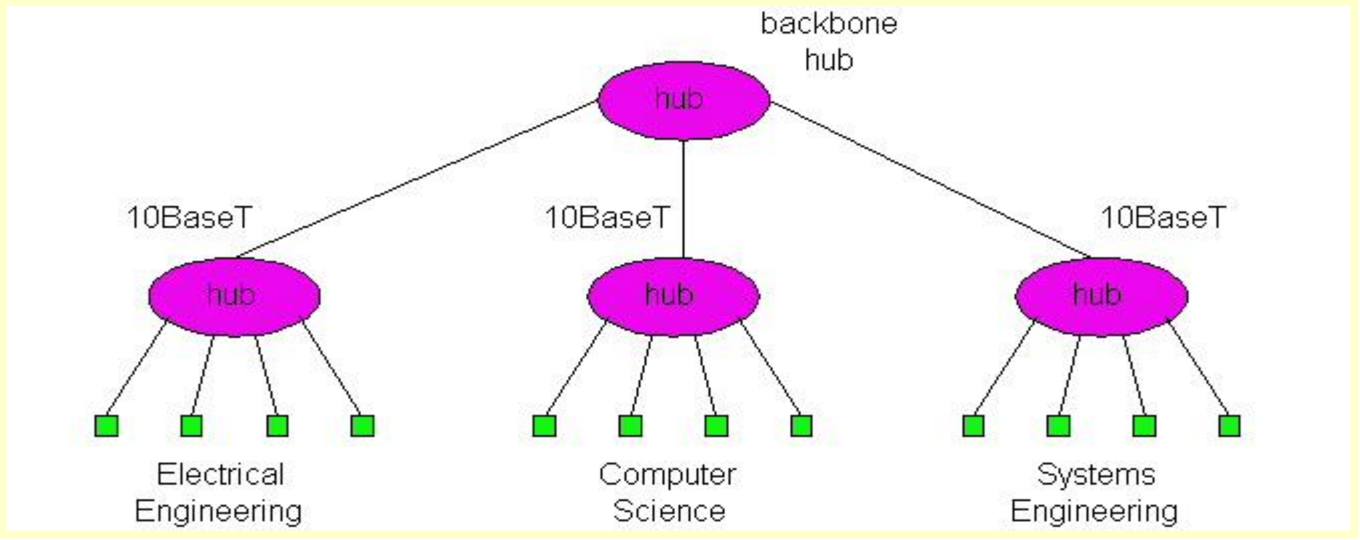
- **10: 10Mbps; 2: under 200 meters max cable length**
- **Thin coaxial cable in a bus topology**
- **MAX 30 users on one segment.**



- **Repeaters used to connect up to multiple segments**
- **Repeater repeats bits it hears on one interface to its other interfaces: physical layer device!**

10BaseT and 100BaseT - 1

- **10/100 Mbps rate; latter called "fast ethernet"**
- **T stands for Twisted Pair**
- **Hub to which nodes are connected by twisted pair, thus "star topology"**
- **CSMA/CD implemented at hub**



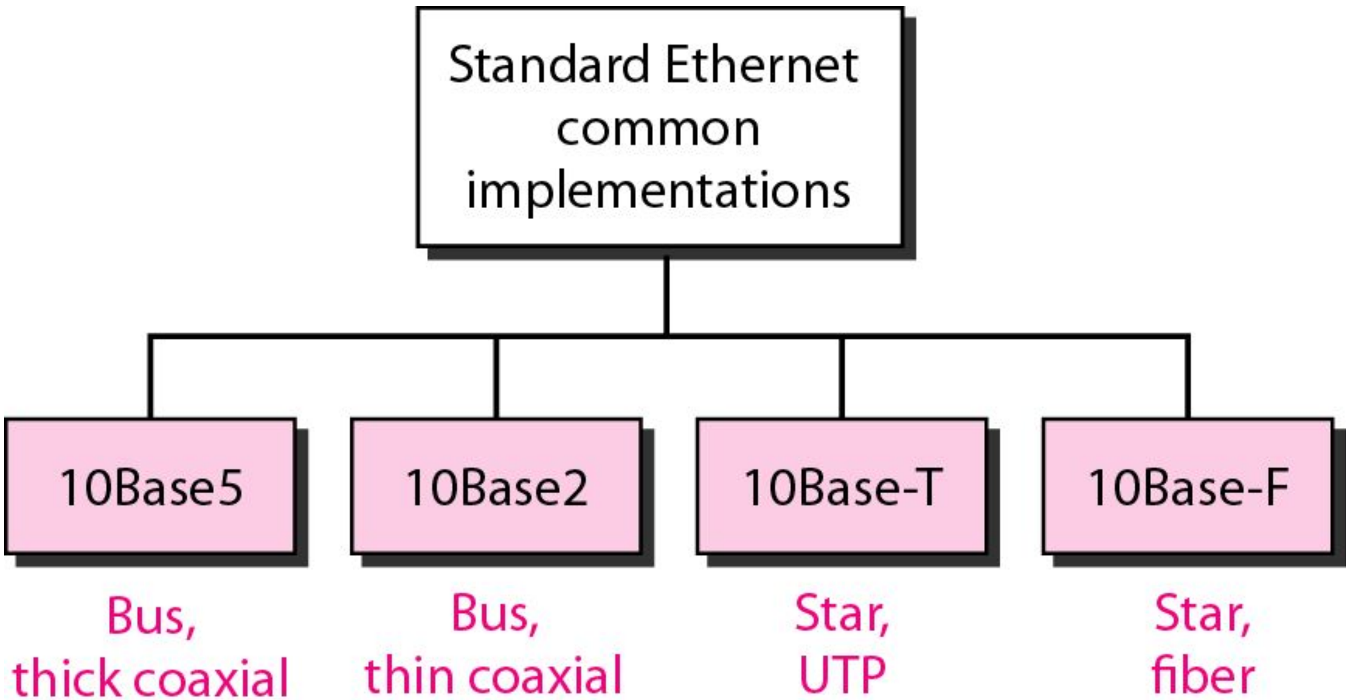
10BaseT and 100BaseT - 1

- Max distance from node to Hub is 100 meters
- Hub can disconnect “jabbering adapter
- Hub can gather monitoring information, statistics for display to LAN administrators

Gbit Ethernet

- Use standard Ethernet frame format
- Allows for point-to-point links and shared broadcast channels
- In shared mode, CSMA/CD is used; short distances between nodes to be efficient
- Uses hubs, called here “Buffered Distributors”
- Full-Duplex at 1 Gbps for point-to-point links

Categories of Standard Ethernet



Summary of Standard Ethernet implementations

Characteristics	*10Base5*	*10Base2*	*10Base-T*	*10Base-F*
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester